

Chapter 5 Homework: The Multiple Regression Model

Claire

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Exercise 5.5

For each of the following time-series regression models, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

Part (a): $y_t = \beta_1 + \beta_2 t + e_t, \quad t = 1, 2, \dots, T$

$$\text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\sum_{t=1}^T (t - \bar{t})^2}$$

Using the identities:

$$\sum_{t=1}^T t = \frac{T(T+1)}{2}, \quad \sum_{t=1}^T t^2 = \frac{T(T+1)(2T+1)}{6}$$

$$\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - T\bar{t}^2 = \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} = \frac{T(T^2-1)}{12}$$

Therefore:

$$\text{var}(b_2|\mathbf{x}) = \frac{12\sigma^2}{T(T^2-1)}$$

```
# Verify numerically for large T
var_b2_a <- function(T, sigma2 = 1) {
  t_vals <- 1:T
  tbar <- mean(t_vals)
  Sxx <- sum((t_vals - tbar)^2)
  sigma2 / Sxx
}
cat("var(b2) at T=10:", round(var_b2_a(10), 6), "\n")
```

```
## var(b2) at T=10: 0.012121
```

```
cat("var(b2) at T=100:", round(var_b2_a(100), 8), "\n")
```

```
## var(b2) at T=100: 1.2e-05
```

```
cat("var(b2) at T=1000:", round(var_b2_a(1000), 12), "\n")
```

```
## var(b2) at T=1000: 1.2e-08
```

Conclusion for (a): As $T \rightarrow \infty$, $\text{var}(b_2|\mathbf{x}) = \frac{12\sigma^2}{T(T^2-1)} \rightarrow 0$. The OLS estimator is **consistent**.

Part (b):

$$y_t = \beta_1 + \beta_2(0.5)^t + e_t, \quad t = 1, 2, \dots, T$$

The regressor is $x_t = (0.5)^t$. Using the geometric series formula:

$$\sum_{t=1}^T x_t = \sum_{t=1}^T (0.5)^t = \frac{0.5(1 - 0.5^T)}{1 - 0.5} = 1 - (0.5)^T \rightarrow 1 \text{ as } T \rightarrow \infty$$

$$\sum_{t=1}^T x_t^2 = \sum_{t=1}^T (0.25)^t = \frac{0.25(1 - 0.25^T)}{0.75} = \frac{1}{3}(1 - 0.25^T) \rightarrow \frac{1}{3}$$

$$\bar{x} = \frac{1}{T} \sum x_t \rightarrow \frac{1}{T} \rightarrow 0, \quad \sum (x_t - \bar{x})^2 \rightarrow \frac{1}{3} - 0 = \frac{1}{3}$$

```
var_b2_b <- function(T, sigma2 = 1) {
  t_vals <- 1:T
  x_vals <- (0.5)^t_vals
  xbar <- mean(x_vals)
  Sxx <- sum((x_vals - xbar)^2)
  sigma2 / Sxx
}
cat("var(b2) at T=10:", round(var_b2_b(10), 6), "\n")
```

```
## var(b2) at T=10: 4.282137
```

```
cat("var(b2) at T=100:", round(var_b2_b(100), 6), "\n")
```

```
## var(b2) at T=100: 3.092784
```

```
cat("var(b2) at T=1000:", round(var_b2_b(1000), 6), "\n")
```

```
## var(b2) at T=1000: 3.009027
```

Conclusion for (b): As $T \rightarrow \infty$, $\text{var}(b_2|\mathbf{x}) \rightarrow \frac{\sigma^2}{1/3} = 3\sigma^2 > 0$. The variance **does not converge to zero**, so b_2 is **not consistent**.

Part (c): Intuitive Explanation

- In **(a)**, $x_t = t$ grows without bound, providing increasing variation in the regressor. The sum of squared deviations grows as $T^3/12$, swamping the numerator σ^2 , driving the variance to zero.
- In **(b)**, $x_t = (0.5)^t$ decays exponentially to zero. The regressor converges, so its variation saturates at a finite level. No matter how many observations we add, we gain essentially no new information about β_2 , and the variance remains positive.

Exercise 5.6

From a sample of $N = 63$ observations, the least squares estimates and the estimated covariance matrix are:

$$\mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}, \quad \widehat{\text{cov}}(\mathbf{b}) = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{pmatrix}$$

```
b <- c(2, 3, -1)
cov_b <- matrix(c(3, -2, 1,
                  -2, 4, 0,
                  1, 0, 3), nrow = 3, byrow = TRUE)

N <- 63
K <- 3      # number of parameters (including intercept)
df <- N - K

alpha <- 0.05
t_crit <- qt(1 - alpha/2, df = df)
cat("t-critical (5%, df=60):", round(t_crit, 4), "\n")
```

```
## t-critical (5%, df=60): 2.0003
```

Part (a): Test $H_0 : \beta_2 = 0$ vs $H_1 : \beta_2 \neq 0$

```
# H0: beta2 = 0
c_vec <- c(0, 1, 0) # contrast vector for beta2
est <- sum(c_vec * b)
se <- sqrt(t(c_vec) %*% cov_b %*% c_vec)
t_stat <- est / se

cat("Estimate (b2):", est, "\n")
```

```
## Estimate (b2): 3
```

```
cat("SE:", round(se, 4), "\n")
```

```
## SE: 2
```

```
cat("t-statistic:", round(t_stat, 4), "\n")
```

```
## t-statistic: 1.5
```

```
cat("t-critical:", round(t_crit, 4), "\n")
```

```
## t-critical: 2.0003
```

```
cat("Reject H0?", abs(t_stat) > t_crit, "\n")
```

```
## Reject H0? FALSE
```

Conclusion: Since $|t| = 1.5 < t_{0.025,60} = 2$, we **fail to reject** H_0 at the 5% level.

Part (b): Test $H_0 : \beta_1 + 2\beta_2 = 5$ vs $H_1 : \beta_1 + 2\beta_2 \neq 5$

```
c_vec2 <- c(1, 2, 0)
r      <- 5
est2   <- sum(c_vec2 * b)
se2    <- sqrt(t(c_vec2) %*% cov_b %*% c_vec2)
t_stat2 <- (est2 - r) / se2

cat("Estimate (b1 + 2*b2):", est2, "\n")
```

```
## Estimate (b1 + 2*b2): 8
```

```
cat("SE:", round(se2, 4), "\n")
```

```
## SE: 3.3166
```

```
cat("t-statistic:", round(t_stat2, 4), "\n")
```

```
## t-statistic: 0.9045
```

```
cat("Reject H0?", abs(t_stat2) > t_crit, "\n")
```

```
## Reject H0? FALSE
```

Part (c): Test $H_0 : \beta_1 - \beta_2 + \beta_3 = 4$ vs $H_1 : \beta_1 - \beta_2 + \beta_3 \neq 4$

```
c_vec3 <- c(1, -1, 1)
r3     <- 4
est3   <- sum(c_vec3 * b)
se3    <- sqrt(t(c_vec3) %*% cov_b %*% c_vec3)
t_stat3 <- (est3 - r3) / se3

cat("Estimate (b1 - b2 + b3):", est3, "\n")
```

```
## Estimate (b1 - b2 + b3): -2
```

```
cat("SE:", round(se3, 4), "\n")
```

```
## SE: 4
```

```
cat("t-statistic:", round(t_stat3, 4), "\n")
```

```
## t-statistic: -1.5
```

```
cat("Reject H0?", abs(t_stat3) > t_crit, "\n")
```

```
## Reject H0? FALSE
```

Exercise 5.20

A generalized version of the estimator for β_2 proposed by Professor I.M. Mean:

$$\hat{\beta}_{2,\text{mean}} = \frac{y_2 - y_1}{x_2 - x_1}$$

where y_1, x_1 are sample means for the first half and y_2, x_2 for the second half of observations (ordered by increasing x).

Part (a): Show $\hat{\beta}_{2,\text{mean}}$ is unbiased

Under MR1–MR6, $y_i = \beta_1 + \beta_2 x_i + e_i$. Let the first half have indices $i = 1, \dots, N/2$ and the second half $i = N/2 + 1, \dots, N$.

$$y_j = \beta_1 + \beta_2 x_j + e_j, \quad j = 1, 2$$

$$\hat{\beta}_{2,\text{mean}} = \frac{(\beta_1 + \beta_2 x_2 + e_2) - (\beta_1 + \beta_2 x_1 + e_1)}{x_2 - x_1} = \beta_2 + \frac{e_2 - e_1}{x_2 - x_1}$$

Taking the conditional expectation (given $\mathbf{x}$):

$$E(\hat{\beta}_{2,\text{mean}}|\mathbf{x}) = \beta_2 + \frac{E(e_2|\mathbf{x}) - E(e_1|\mathbf{x})}{x_2 - x_1} = \beta_2 + 0 = \beta_2$$

since $E(e_i|\mathbf{x}) = 0$ by MR2. Thus $\hat{\beta}_{2,\text{mean}}$ is **unbiased**.

Part (b): Derive $\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{x})$

$$\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{x}) = \frac{\text{var}(e_2 - e_1|\mathbf{x})}{(x_2 - x_1)^2}$$

Since e_i 's are uncorrelated with variance σ^2 :

$$\text{var}(e_j|\mathbf{x}) = \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N}, \quad \text{cov}(e_1, e_2|\mathbf{x}) = 0$$

$$\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{x}) = \frac{\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}}{(x_2 - x_1)^2} = \frac{4\sigma^2}{N(x_2 - x_1)^2}$$

```
# Numerical demonstration
set.seed(42)
N <- 100
x <- sort(runif(N))
e <- rnorm(N, sd = 1)
b1_true <- 2; b2_true <- 3
y <- b1_true + b2_true * x + e

# Split halves
x1 <- x[1:(N/2)]; x2 <- x[(N/2+1):N]
y1 <- y[1:(N/2)]; y2 <- y[(N/2+1):N]
beta_mean <- (mean(y2) - mean(y1)) / (mean(x2) - mean(x1))

cat("OLS estimate of beta2: ", round(coef(lm(y ~ x))[2], 4), "\n")
```

```
## OLS estimate of beta2: 2.5198
```

```
cat("Mean estimator (beta_mean):", round(beta_mean, 4), "\n")
```

```
## Mean estimator (beta_mean): 2.5163
```

```
cat("True beta2:", b2_true, "\n")
```

```
## True beta2: 3
```

```
# Compare variances from many simulations
sims <- replicate(2000, {
  e_s <- rnorm(N)
  y_s <- b1_true + b2_true * x + e_s
  ols <- unname(coef(lm(y_s ~ x))[2]) # unname() prevents "ols.x" row label
  y1s <- y_s[1:(N/2)]; y2s <- y_s[(N/2+1):N]
  mean_est <- (mean(y2s) - mean(y1s)) / (mean(x2) - mean(x1))
  c(ols = ols, mean_est = mean_est)
})

cat("\nSimulated variance - OLS: ", round(var(sims["ols",]), 6), "\n")
```

```
##
## Simulated variance - OLS: 0.109674
```

```
cat("Simulated variance - Mean est.: ", round(var(sims["mean_est",]), 6), "\n")
```

```
## Simulated variance - Mean est.: 0.14697
```

```
cat("(OLS has smaller variance by Gauss-Markov)\n")
```

```
## (OLS has smaller variance by Gauss-Markov)
```

Conclusion: $\hat{\beta}_{2,\text{mean}}$ is unbiased but has larger variance than the OLS estimator. By the Gauss-Markov theorem, OLS is BLUE, so $\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{x}) \geq \text{var}(b_2|\mathbf{x})$.